

Spatial Interaction in HRI

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Importance of spatial placement

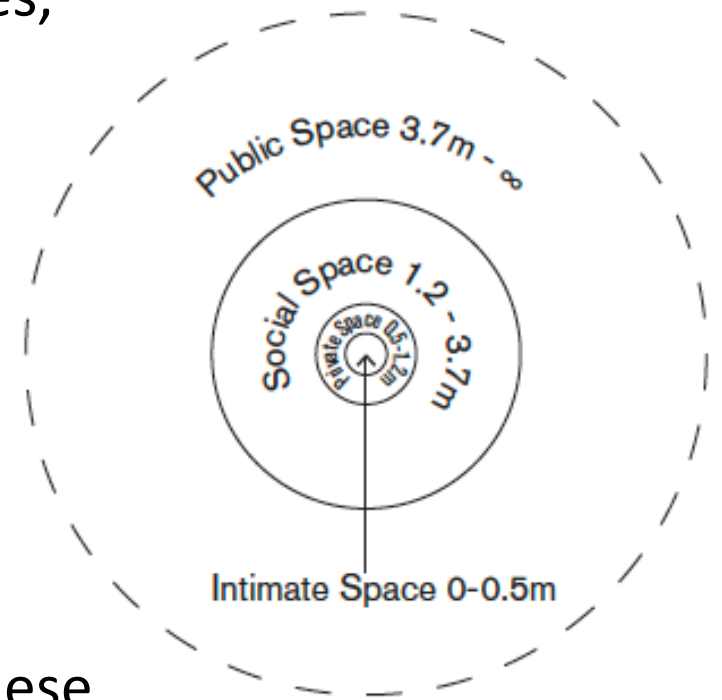
- The placement of the robot with respect to the user is an important aspect of HRI.
- Where should an exercise companion drone be placed?
 - In front of the runner was the best position for the **Joggobot**.
 - Having the **drone flying behind** the jogger made people feel like they were **being chased**, decreasing their enjoyment.
- When **only taking the need for collision avoidance** into account while deciding on the optimal location or path of a robot, one might inadvertently create robot behavior that is considered **uncomfortable, rude, or inappropriate**.



The Joggobot drone

Use of space in human interaction: Proxemics

- **Proxemics** describe how people take up space in relation to others and how spatial positioning influences attitudes, behaviors, and interpersonal interaction.
- Hall et al. (1968) described **four distance zones**:
 1. intimate distance,
 2. personal distance,
 3. social distance, and
 4. public distance
- When the **available space is (relatively) unlimited**, these distances indicate the **psychological closeness** between people.



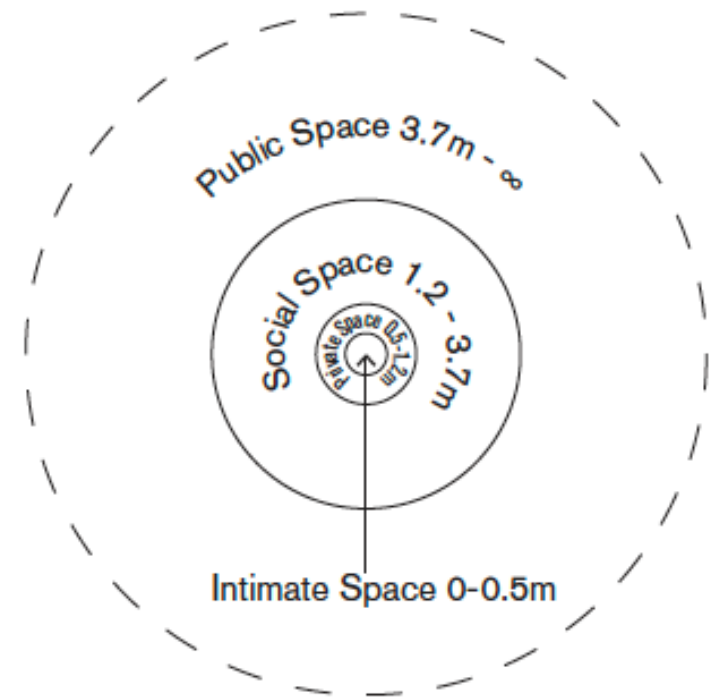
Use of space in human interaction: Proxemics

1. Intimate distance

- Ranges roughly from a few centimeters to about half a meter, depending on one's age and culture.
- Reserved for close personal relationships or the sharing of private information.

2. Personal distance

- Ranges from about half a meter to 1.2 meters.
- Intimate and personal distances together make **up the personal space of a person**: the amount of space that people generally consider theirs to take up.
- Under normal circumstances, **only friends, relatives, and partners** are expected to come this close.



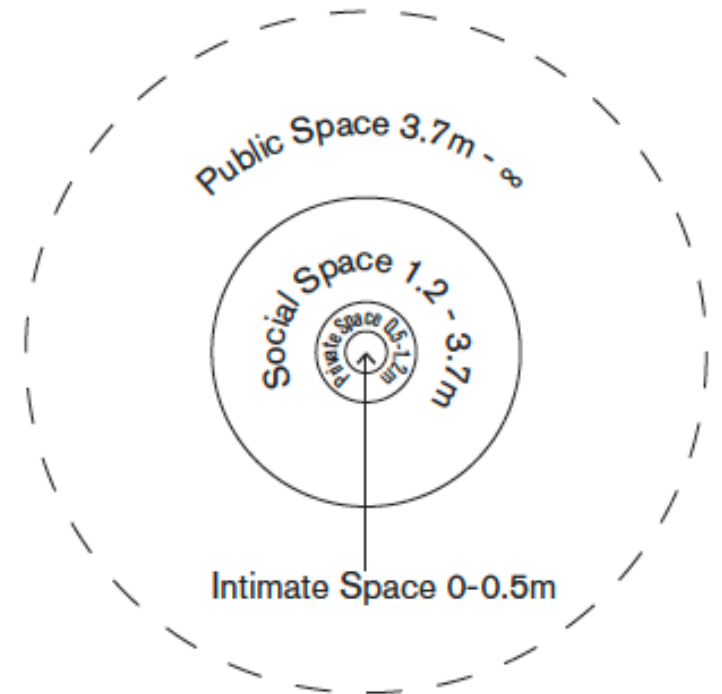
Use of space in human interaction: Proxemics

3. Social distance

- Ranges between 1.2 and about 4 meters.
- Applicable for less personal relationships, such as acquaintances or colleagues.

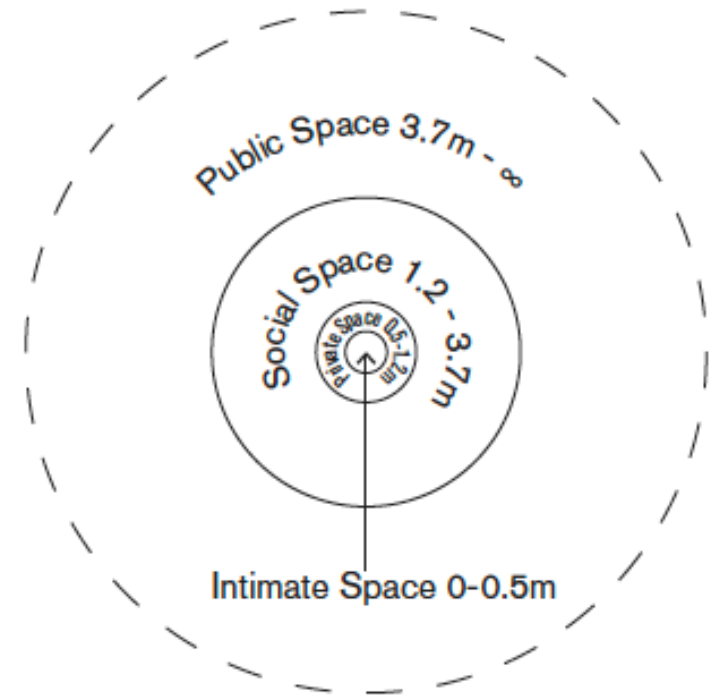
4. Public distance

- Starts at around 4 meters, which is the distance people are expected to keep between them in relatively impersonal settings, such as public speaking at a conference.



Use of space in human interaction: Proxemics

- Cultural impact
 - People from different cultures have varying personal proxemic preferences.
 - For example,
 - In “**high-contact cultures**” such as those of South America, people will frequently enter each other's personal space and touch.
 - Whereas in “**low-contact cultures**”, such as the United States, touching a stranger may be construed as assault.



❖ H. W.: Categorize major countries across different continents based on high- or low-contact cultures?

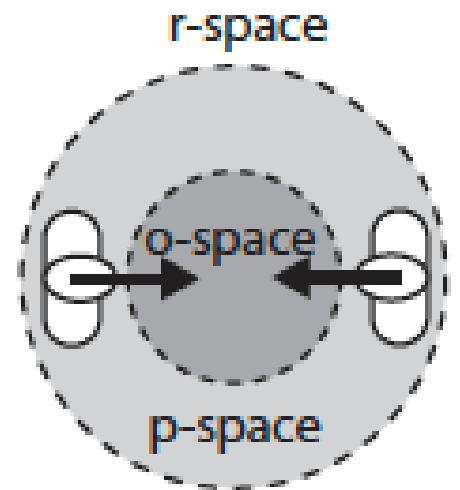
Use of space in human interaction: Group spatial interaction dynamics

- The **spatial orientation of people in a group** can make the group seem as if it were inviting more members or seeking to keep others out.
- For example, at a cocktail party, when people **stand in a tight-knit circle**, it can seem difficult to join in the conversation.
- However, if the group notices people wanting to join and **opens up the circle**, it can be construed as an invitation to participate.
- This type of information can be **useful for robots to gauge** which groups of people they can approach in public spaces like museums or malls.



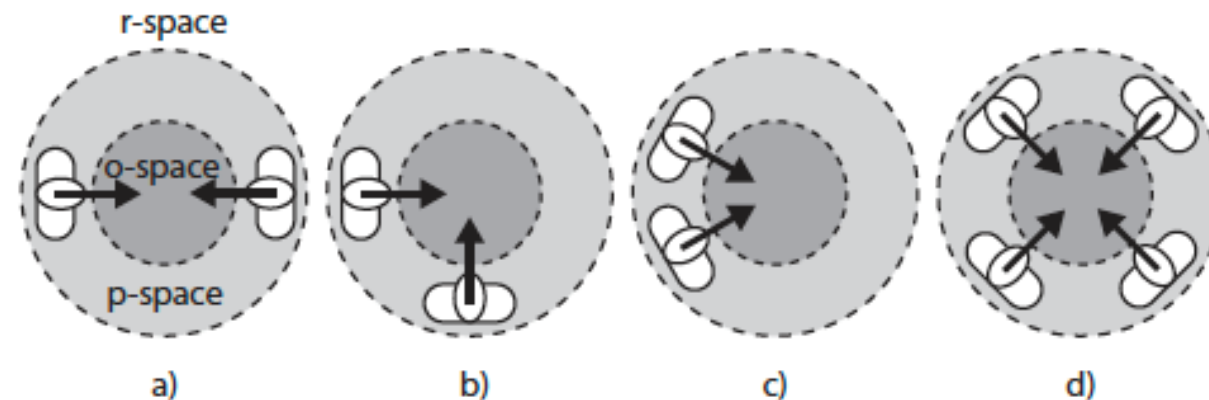
Use of space in human interaction: Group spatial interaction dynamics

- Group spatial dynamics were described by **Adam Kendon** as the “facing formation,” or “F-formation”, which is created through the **positioning of two or more people in relation to each other**, such that the areas of space that they are facing and on which they focus their attention are overlapping.
- The space between these people, which is “one to which they have equal, direct, and exclusive access,” is termed the **o-space**.
- The group participants themselves are said to occupy the **p-space**, and they are surrounded by **r-space**.



Use of space in human interaction: Group spatial interaction dynamics

- Different configurations of the F-formation are possible, based on people's orientation to each other, and are termed the:
 - a) face-to-face,
 - b) L-shape,
 - c) side-by-side formation for two people, and
 - d) the circular formation and other shapes for larger groups.

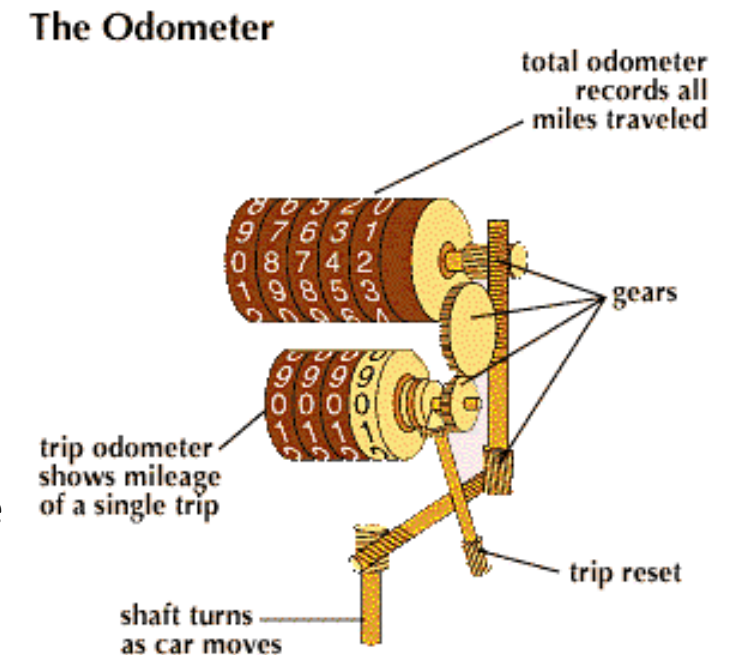


Spatial interaction for robots

- Robots that **do not respect the personal space** of the user will evoke **negative reactions** from the user.
- Therefore, robot designers can attempt to **increase the acceptance** of the robot by having it keep an appropriate distance and by adjusting its position.
- **Four factors** to consider in this regard-
 1. Localization and navigation,
 2. Socially appropriate positioning,
 3. Spatial dynamics of initiating HRI, and
 4. Informing users of the robot's intent.

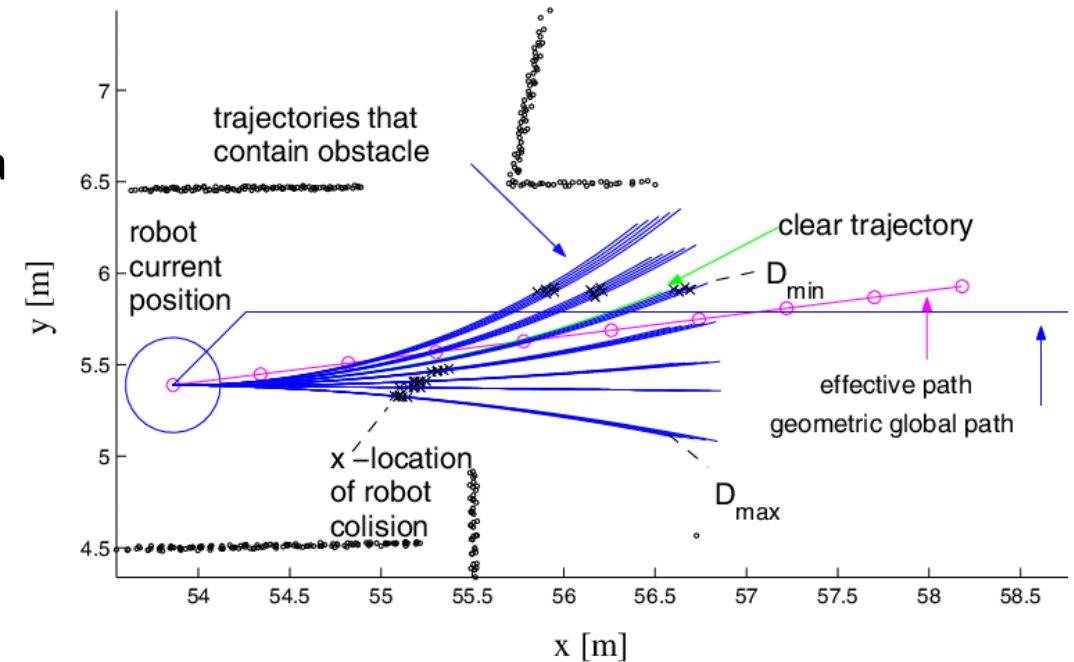
Spatial interaction for robots: Localization and navigation

- Localization:
 - The process of a robot to **locate itself in space** with regard to the people it aims to interact with.
 - A typical robot is equipped with an **odometer**, a sensor that records the distance travelled by the robot's wheels.
 - However, as the robot travels, these **lose accuracy**, and the robot, therefore, needs to correct the information that the odometry provides about its location.
 - The **typical solution** to this is to let the robot build a map of its environment and then cross-reference information on its location and orientation from the odometry with information from other sensors, such as a laser range finder or **camera**.
 - This process is known as **simultaneous localization and mapping, or SLAM**.



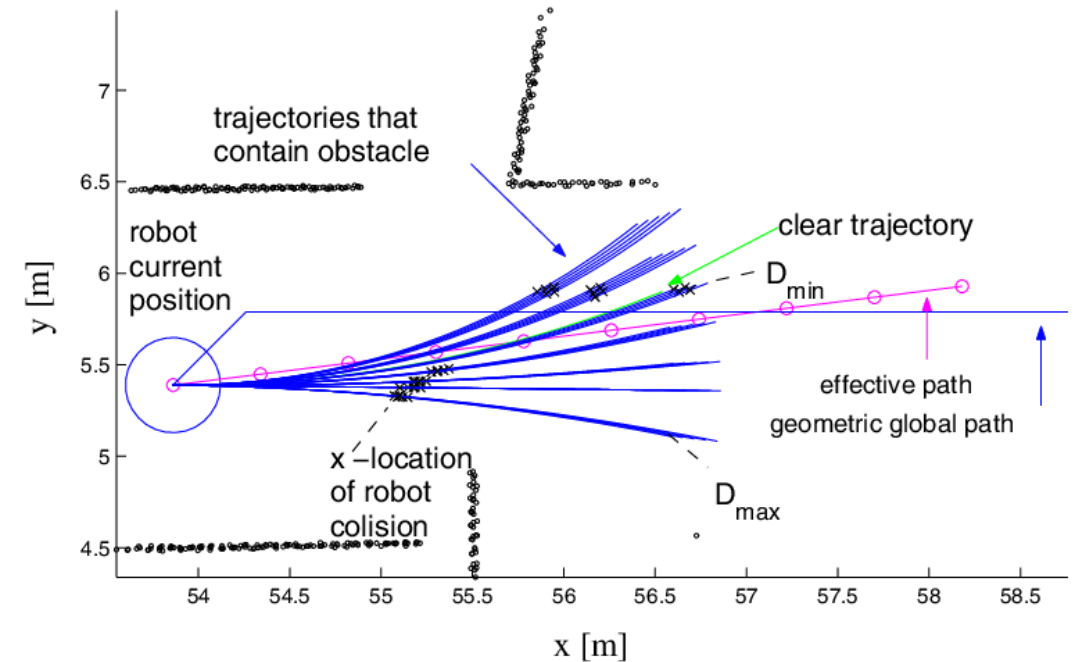
Spatial interaction for robots: Localization and navigation

- Navigation:
 - Moving the robot through a crowded environment is known as **robot navigation**.
 - To avoid collisions between the robot and objects or people, techniques such as the **dynamic window approach (DWA)** are often used. The idea behind this technique is that
 - a system **computes its future location based on the current velocity** of the robot,
 - while at the same time **considering whether to keep or alter its velocity** within the limitations of its capability, and
 - while **calculating a future velocity** that does not result in a collision.



Spatial interaction for robots: Localization and navigation

- Navigation:
 - Over longer time scales, there are techniques based on **path planning**.
 - In these techniques, if a given goal of a robot is not within immediate view of the robot, **a path planning algorithm computes a set of waypoints or paths for the robot that will let it reach its goal**.
 - In HRI, most path-planning algorithms that work well for navigating around obstacles will result in **socially inappropriate behavior** when tried around people.



Spatial interaction for robots: Socially appropriate positioning

- One of the focuses in investigating proxemics in HRI has been identifying appropriate interaction distances between users and robots.
- These include questions like the following:
 - How close do people prefer to stand relative to a robot?
 - How close should a robot approach people before it is considered rude or inappropriate or makes people feel uncomfortable?
- Walters et al. (2005) reported that the majority of people prefer a personal or social distance when interacting with a robot, although some people prefer to stand even closer.
- Huttenrauch et al. (2006) reported that people preferred that the robot stand at distances derived from human proxemics.
- Kuzuoka et al. (2010) reported that a robot can change the conversational F-formations of the group by changing its body orientation, and they also found that movement of the robot's whole body was more effective than having the robot just move its head.

Spatial interaction for robots: Socially appropriate positioning

- **Relational position** is also important when people and robots interact while they are moving.
 - For instance, **when a robot follows a user from behind**, the robot can either follow the **same trajectory** as the user, or it can **move directly to the user's current location**, which might be a shorter and faster pathway. Gockley et al. (2007) showed that users perceive the **first behavior as more natural**.
 - Morales Saiki et al. (2012) developed a technique that allows a robot to navigate side by side with its user, for which they found it important for the robot to anticipate the user's future motion.
 - Considerations of **comfort and perceived safety** should also be integrated into path planning.

Spatial interaction for robots: Spatial dynamics of initiating HRI

- Imagine the moment when a person meets a robot for the first time:
 - Should the robot initiate the interaction?
- In the case where a person approaches a robot, the robot should respond at just the right moment.
- In a situation where a robot joins a queue, the robot needs to respect the personal space of other people who are also waiting (Nakauchi and Simmons, 2002).
- When a robot encounters people, it needs to switch its navigation mode from purely functional to considering social distance and spatial configuration (Althaus et al., 2004).

Spatial interaction for robots: Spatial dynamics of initiating HRI

- Social navigation has become particularly relevant in the context of self-driving cars.
 - The story goes that the **first self-driving cars at Google** drove optimal trajectories following the highway code, but they frequently startled other road users by driving too close or cutting them off.
 - Only when **politeness** was explicitly added as an optimization criterion did the cars drive in a socially acceptable way.

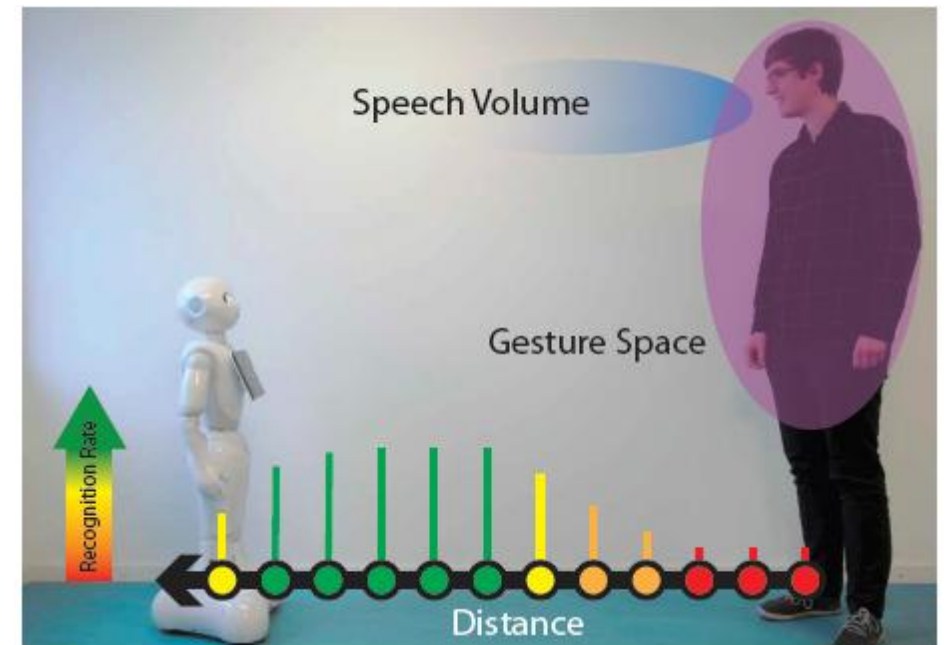
Spatial interaction for robots: Informing users of the robot's intent

- Robot motion trajectories are often used to convey the intent and goal of the robot.
- Path-planning algorithms have been developed to explicitly convey information through the robot's trajectory.
 - For instance, by slowly passing a few meters from a visitor, a mobile robot is able to express whether it is available for an interaction.
- During handover in HRI, users prefer a robot to behave with “legibility”- in a way that allows users to understand its goal and intention.
 - A robot could hand over an object to a person in many different ways, but the most energy-efficient way may be incomprehensible to a person, so it is better to perform a motion that is easier to interpret.

❖ Think about the use of a robot in various daily situations (e.g., at home, at the office, and on a crowded train). Now, think about how you need to adapt the spatial-navigation behavior of the robot to each of these contexts. What would be important factors to consider in these different contexts?

Conclusion

- The study of spatial interaction in HRI is often inspired by our understanding of
 - human proxemics,
 - conversational relations, and
 - relational positioning and approach behaviors.
- However, **norms and understandings that are common knowledge for people**, to the point where they may not even be aware of them anymore, often turn out to be **not so trivial** to incorporate into robot behavior.



THE END